Question 1

Not answered

Marked out of 1.00

Flag question

Find

$$(11010 \wedge 10011) \oplus 01000$$

10010 01000

Answer: 11010



Question 2

Not answered

Marked out of 1.00

Flag question

Let *p*, *q* be propositions:

p = "You do all homework problems"

***q* = "You receive good grades"**

Translate the sentence into logical expression:

"Doing all homework problems is enough to receive good grades".

Select one:

- a. $p \vee q$
- b. $q \rightarrow p$
- c. $p \wedge q$
- d. $p \rightarrow q$
- e. Các lựa chọn còn lại đều sai

Question 3

Not answered

Marked out of 1.00

Flag question

Let p, q be two propositions. Which propositions are logically equivalent to $\neg p \oplus q$?

1

1 1

$(\neg p \vee q) \vee (p \vee \neg q)$ Choose... false

1 1

$(\neg p \vee q) \wedge (p \vee \neg q)$ Choose... True

1 1

$(\neg p \vee q) \oplus (p \vee \neg q)$ Choose... False

Question 4

Not answered

Marked out of 1.00

Flag question

Which propositions are contradiction?

$[(p \rightarrow q) \vee (q \rightarrow p)] \wedge (p \oplus q)$ Choose... false

$(p \rightarrow q) \wedge (q \rightarrow p) \wedge (p \oplus q)$ Choose... true

$(p \rightarrow q) \vee (q \rightarrow p) \vee (p \oplus q)$ Choose... false

Question 5

Not answered

Marked out of 1.00

Flag question

Which statements are correct?

$\forall x(P(x) \wedge Q(x))$ and $\forall x P(x) \wedge \forall x Q(x)$ have the same truth values — true — Choos

$\forall x(P(x) \rightarrow Q(x))$ and $\forall x P(x) \rightarrow \forall x Q(x)$ have the same truth values — false — Choos

$\forall x(P(x) \vee Q(x))$ and $\forall x P(x) \vee \forall x Q(x)$ have the same truth values — flase — Choos

Question 6

Not answered

Marked out of 1.00

Flag question

Translate the logical expression into sentence, domain is all real numbers

$$\forall x \forall y((xy = 0) \rightarrow ((x = 0) \vee (y = 0)))$$

Select one:

- a. The product of 0 and 0 is 0
- b. If x = 0 then for all y we have xy = 0
- c. If the product of two numbers is 0 then at least one of them is 0
- d. If the product of two numbers is 0 then both of them are 0

Question 7

Not answered

Marked out of 1.00

Flag question

Given the hypotheses:

- **Every FPT students stays in the dorm.**
- **An is staying in the dorm.**
- **Bình is not staying in the dorm.**

Which conclusion can be drawn?

Select one:

- a. An is a student of FPT
- b. Bình is not a student of FPT, and An is.
- c. Bình is a student of FPT, and An is not.
- d. Bình is not a student of FPT

Question 8

Not answered

Marked out of 1.00

Flag question

Find the negation of

$\exists x \exists y (\forall z T(x,y,z) \vee Q(x,y))$ **?**

Select one:

a. $\forall x \forall y (\exists z \neg T(x,y,z) \wedge \neg Q(x,y))$

b. $\forall x \exists y (\forall z \neg T(x,y,z) \wedge \neg Q(x,y))$

c. $\forall x \forall y (\exists z T(x,y,z) \wedge Q(x,y))$

d. $\forall x \forall y (\exists z \neg T(x,y,z) \vee \neg Q(x,y))$

Question 9

Not answered

Marked out of 1.00

Flag question

Determine if each statement is true of false.

$A - \emptyset = \emptyset$ Choose... false

$A \cup \emptyset = \emptyset$ Choose... false

$\emptyset - A = \emptyset$ Choose... true

$A \cap A = A$ Choose... true

Question
10

Not answered

Marked out of 1.00

Flag question

Find the cardinality of the set { 1, 2, 3, 2, 5, 6 }

Answer: 5



Question
11

Not answered

Marked out of 1.00

Flag question

Compute $\left\lceil \left(\frac{7}{2}\right)^2 \right\rceil - \left(\left\lceil \frac{7}{2} \right\rceil\right)^2$

Answer: -3

Question
12

Not answered

Marked out of 1.00

Flag question

Let f(X) = 5X + 4, g(X) = 4X + 3. Suppose that f o g (X) = aX + b.

Find a + b.

$5(4x+3)+4$

$20x+15+4$

Answer: 39



Question
13

Not answered

Marked out of 1.00

Flag question

Compute $\left\lfloor \frac{3}{2} - \left\lceil 3 + \frac{5}{4} \right\rceil \right\rfloor$

Answer: -3

Question
14

Not answered

Marked out of 1.00

Flag question

Let f: Z x Z --> Z, f(m, n) = n+1. Choose correct answer:

Select one:

- a. f is one-to-one but not onto
- b. f is onto but not one-to-one
- c. f(x) is neither one-to-one nor onto
- d. f is a bijection



Question
15

Not answered

Marked out of 1.00

Flag question

Compute

$$\sum_{i=0}^{10} (2^{i+1} - 2^i).$$

Answer: 2047